Agenthon Track 2 — Phase 2 results

Measured October 14, 2026 · 2,000 draws · Lower scores are better

Verdict

The frozen numeric_v1 improves the development CV and reserved-year means, and passes all 104 admissibility checks. **The paired capped-card reduction target is unmet.** It ties M0's total cap count and increases caps in 2019–2024. The final mean improvement is small relative to year-to-year uncertainty, so these results do not establish a robust improvement in the sealed final window. No candidate was tuned after revealing reserved-year or practice results.

Main results

Evaluation	Cards	M0 mean	Numeric mean	M0 caps	Numeric caps	Numeric median
Primary year CV: 2009–2018	1073	1.033389	0.953077	1	1	0.703835
Reserved: 2019–2024	554	1.480838	1.456363	15	17	0.808653
All historical, descriptive	2060	1.210785	1.154244	23	23	0.760472
Practice proxies, report only	88	2.551735*	2.435614	11*	9	1.376961

*Practice M0 is the original 500-draw measurement; all historical paired rows recompute M0 at 2,000 draws. Original historical M0 was 1.214629 with 24 caps; its 2,000-draw control is 1.210785 with 23. Comparing 23 numeric caps against 24 original M0 caps would confound model and Monte Carlo changes.

Reserved-year wins: **4 of 6**. The equal-year mean paired delta is **–0.034823**, with year-cluster standard error **0.050554** (only six years). This uncertainty includes zero improvement. The primary CV winner was selected using the same development CV scores, so its CV result may be optimistic. Reserved monthly macro also worsens from **1.6288 to 2.0143 (+23.7%)** on 26 cards.

Validation protocol

Step 1 diagnosis was written to BUILD_LOG before Step 2; the protocol was recorded before any candidate score. Development uses 2005–2018, with 2005–2008 warm-up labels and one primary test fold for each year 2009–2018 (1,073 cards). Fixed rules estimate moments only from each card's pre-cutoff history. The only outcome-fitted parameter is global width; each fold fits it on earlier years, purging cards whose targets cross the test-year boundary. The final rule was frozen before scoring 2019–2024 (554 cards), then practice proxies.

M0 outcomes across all years were inspected in the required diagnosis, so the reserved period is a candidate holdout, not a completely untouched dataset. Practice uses 85 inferred dates and 3 explicit monthly targets. Latest-vintage macro unions, G10 substitutions for EM shapes, and business-day calendars remain limitations. No sealed outcomes or organizer feedback were used.

Ordered ablations

Each row is compared with the previous accepted best, identified in the detailed JSON. Accept only a lower primary mean, no increase in caps, and no >10% loss in a major family/panel/horizon/shape group. Major means at least 40 cards, or 20 for monthly macro. Every candidate and its full breakdowns are retained.

Candidate	CV mean	Median	Caps	Change in mean	Decision
M0 control	1.033389	0.738898	1	+0.000000	Control
Zero drift	0.963807	0.707033	1	-0.069582	Reject: group regression
25% drift	0.959594	0.711342	1	-0.073795	Kept
EWMA volatility	0.960115	0.714071	0	+0.000521	Reject: mean
Recent/history blend	1.025695	0.823174	0	+0.066100	Reject: mean + groups
Empirical variance ratio	0.964567	0.711804	1	+0.004972	Reject: mean
Student-t, df 5	0.999990	0.796998	0	+0.040396	Reject: mean
5% volatility mixture	0.973108	0.744962	0	+0.013514	Reject: mean
Rates mean reversion	0.963709	0.719783	1	+0.004115	Reject: mean
Monthly trend/persistence	0.953506	0.706513	1	-0.006089	Kept
Joint shrinkage/paths	0.953077	0.703835	1	-0.000428	Kept
Fold-fitted width	0.966927	0.725950	1	+0.013850	Reject: mean

The kept rules are 25% trailing drift, monthly positive-index log trend over 12 changes with UNRATE persistence, and 20% correlation shrinkage using up to 1,260 aligned changes. Each draw shares multivariate shocks across assets; later horizons add independent continuation. Volatility remains the original 300-observation estimate, tails remain Gaussian, and width remains 1.0.

Zero drift improved the aggregate but hurt monthly macro beyond the limit. The EWMA and high-volatility alternatives reduced development caps but worsened mean score. Longer-history variance made quiet-period distributions too broad. Rate reversion and empirical variance ratios also failed the mean gate. The joint-stage gain is only 0.000428 and should not be treated as independently statistically established. All tuning menus were fixed in advance.

Reserved-year breakdowns

Group	Cards	M0 mean	Numeric mean	M0 caps	Numeric caps
family: T2-F1	122	2.1836	2.0610	11	12
family: T2-F2	144	1.2691	1.2368	2	1
family: T2-F3	121	1.4725	1.4984	0	2
family: T2-F4	167	1.1561	1.1736	2	2
panel: factors_daily	77	1.1042	1.0850	1	1
panel: g10_fx_daily	183	1.1455	1.0864	2	2
panel: g10_fx_daily+rates_daily	20	1.6292	1.7232	0	1
panel: macro_monthly	26	1.6288	2.0143	2	2
panel: rates_daily	248	1.8177	1.7646	10	11
horizon_bucket: 127+	122	2.1836	2.0610	11	12
horizon_bucket: 1–21	137	1.0789	1.1258	1	1
horizon_bucket: 22–63	158	1.3684	1.3111	0	1
horizon_bucket: 64–126	137	1.3866	1.4161	3	3
shape: joint	236	1.7829	1.7469	10	13
shape: single	318	1.2567	1.2407	5	4
year: 2019	91	1.2568	1.3561	0	0
year: 2020	95	1.5548	1.4570	2	1
year: 2021	110	2.1972	2.0428	13	12
year: 2022	119	1.5860	1.7295	0	4
year: 2023	97	0.8454	0.7542	0	0
year: 2024	42	1.0923	0.9839	0	0

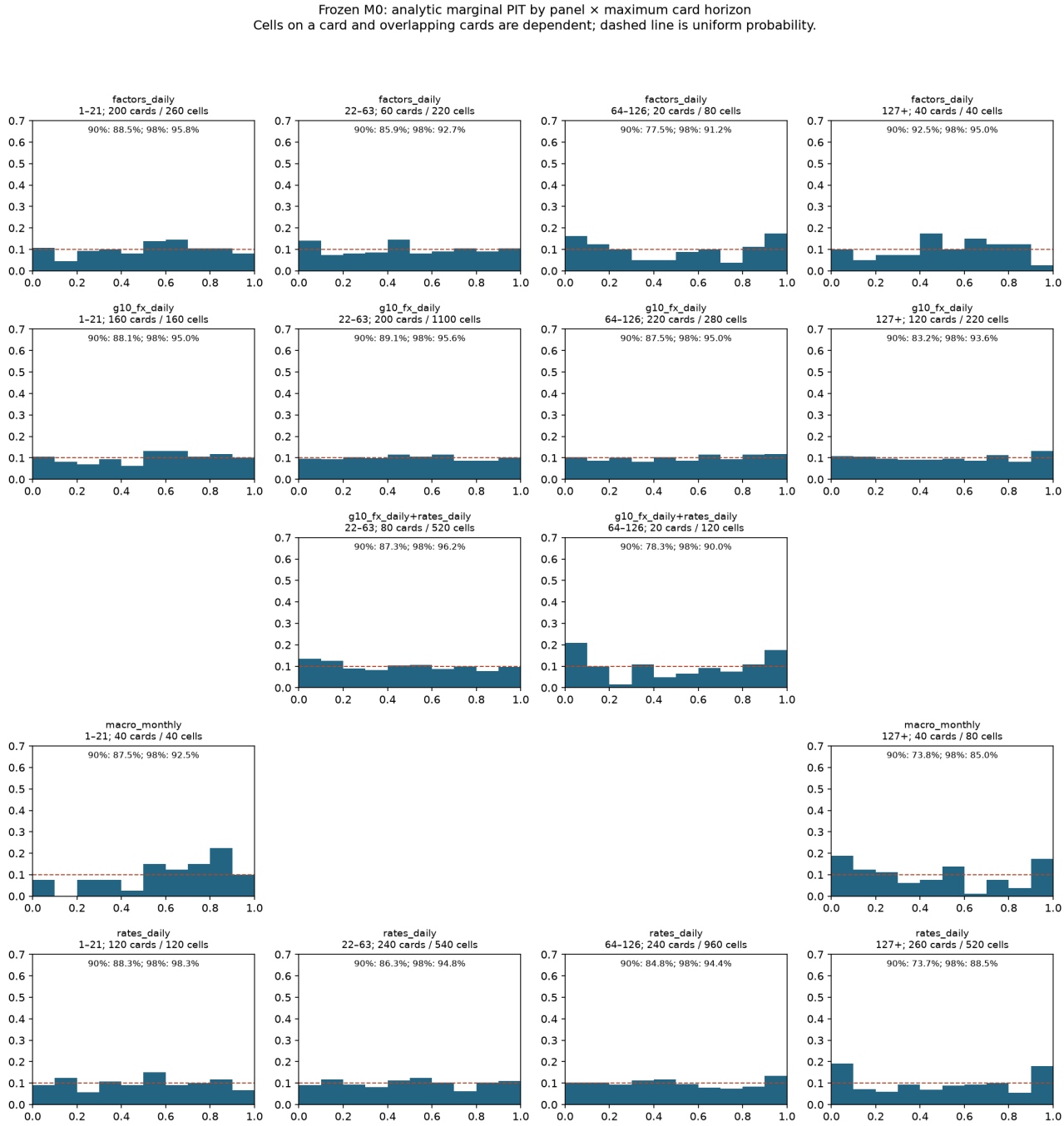
M0 diagnosis before tuning

Nominal 90% and 98% intervals cover **85.13% and 94.09%** of 5,260 cells. Mean z is +0.0193, median +0.0030, RMS 1.3825. Opposing regime errors cancel in the signed average. Long rates and monthly macro are the weakest calibration areas; uniformly doubling width worsens mean score from 1.2146 to 1.5013.

Hindsight center/width experiments allocate 0.7103 of score reduction to center and 0.2572 to width, leaving 0.2471 at the chosen oracle widths. These are postmortem oracle diagnostics, not deployable predictions or irreducible risk. Zero drift alone scores 1.1825 overall but hurts monthly macro.

The original 24 capped cards comprise 15 provisional regime breaks, five monthly/vintage issues, three volatility spikes, and one drift extrapolation. No bug was demonstrated. Repeated UST_2Y cards in late 2021 are overlapping examples of the same quiet-to-moving transition. "Monthly/vintage" flags the known data limitation; it does not prove revisions caused the error.

Full capped-card inventory, all panel × horizon coverage, and methodology are in DIAGNOSIS.md. The CSV files retain all values and PIT bins.



Operational checks

- 52 tests pass, including neutral-hook byte identity, seed determinism, future-data exclusion, independent continuation, monthly mapping, and both fallback levels. Existing macOS NumPy BLAS warnings remain finite-output warnings in baseline tests; numeric_v1 uses explicit NumPy contractions.
- 104/104 official admissibility passes at 2,000 draws, maximum host forecast time 0.066 seconds in the recorded sweep. Historical maximum is 0.094 seconds.
- Full official CLI scoring of 2,060 numeric forecasts agrees with the fast ablation adapter within 8.9e-16. Zero inference failures.
- linux/amd64 build and offline docker-smoke pass g0–g3 with read-only root, no network, and a small non-executable temporary filesystem.
- The image contains only runtime code and numerical dependencies. The build allowlist excludes evaluation data, vendor code, reports and realized values.
- No competition pack or submission upload command was run.

Image publication

The package agenthon-numeric-v1 is **Public**. GitHub Actions completed 52 tests, 104/104 admissibility gates (maximum Linux forecast time 0.174 seconds), an amd64 build, and the offline smoke test before publishing. Successful workflow.

Immutable image:

`ghcr.io/hhwolf/agenthon-numeric-v1@sha256:25ffda44821a0493ba5ba69aeb570ec009a5e1081cd54d5e615fa3d51ac70a01`

The convenience tag is `numeric_v1`; the immutable commit tag is `b1f2225a5ef35a06c6f68e63c4f2c617cf49d300`. The starter-pack anonymous token/manifest check returned HTTP 200, as did all config/layer checks. A real Docker pull passed with an empty temporary credential configuration. The downloaded digest also passed all four official gates offline. The image runs as `runner`, declares no volumes, and contains no evaluation package, vendor tree, or parquet data. Evidence: anonymous-pull.json and published-image-smoke.json.

Artifacts and reproduction

See VALIDATION.md, ablations.json, ablation-breakdowns.csv, frozen-selection.json, and final-evaluation.json. Per-card candidate CSV files include every development/warm-up card; primary means filter 2009–2018. The numeric engine guide gives commands and the optional text-signal contract. BUILD_LOG contains the chronological audit.

Status: the numeric engine and evaluation are implemented. The requested accuracy gate remains open because capped-card count did not improve on the paired historical comparison and increased in the reserved period.